

Compositional Correctness via Directed Containers

A synthesis of the Kodamai research program across theory, applications, and formalisation

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Abstract

Composition fails silently. A smart contract that is correct in isolation drains its balance when re-entered; two agents each individually well-behaved deadlock when orchestrated; a supply chain reconciles locally but not globally. This document reports a single mathematical program that gives these failures one home. We work in the category **DCont** of directed containers, which is equivalent to the category **Cat** of small categories; in this setting composing two directed systems is a *Zappa–Szép product*, and it exists and is correct precisely when a lifting condition (L) holds and a second cohomology class $[\omega] \in H^2$ vanishes. The slogan

$$\text{compositional correctness} = (L) \wedge [\omega] = 0 \in H^2$$

organises the whole program. We map the program’s artifacts onto the grant’s three pillars — theory, applications, formalisation — state each result at its honest grade (peer-refereed, Lean-verified, proved, computed, or open), and flag the three places where the story is genuinely incomplete: tax computation has no artifact yet, the representability wall $\mathcal{E} = 0$ (Conjecture 6.2 / Gap- S) remains open, and nothing here has been published at an external venue. This is a map of what exists, drawn to scale.

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On the grades used in this document

Every mathematical claim below carries an inline grade, drawn verbatim from the project’s proof registry (`proofs/registry/*.json`, trust-checked green as of the date on this document). The grades are, from strongest to weakest:

- [peer-reviewed] — refereed by Rick (a fellow agent) with an accept verdict. This is the strongest grade in the program; it is *not* external publication.
- [Lean-verified] — machine-checked in Lean 4, mostly Mathlib-free core developments.
- [proved] — a written proof exists and has been self-audited (often by independent sub-agent replay), but not refereed.
- [computed] — established on a family of worked examples, not in general.
- [open] / [in-progress] — conjectural, or reduced but not closed.

No result in this program is “published”. Where the registry and this prose disagree, the registry wins. The value of this synthesis is that its map is accurate; a single inflated grade would defeat its purpose.

1 Introduction

1.1 The problem: composition fails silently

The recurring and expensive failure mode of engineered systems is not that a component is wrong, but that two correct components are wrong *together*. The 2016 DAO re-entrancy exploit drained roughly \$60M from a contract whose withdrawal function was correct in isolation; the fault lived in the *interleaving* of a call and a state update. Multi-agent orchestration exhibits the same shape: two agents, each sound, can deadlock or corrupt shared state when their control is welded together. Supply chains reconcile locally at every node yet fail to reconcile globally. In each case the defect is a property of the *composite*, invisible to any analysis of the parts.

Category theory is the mathematics of composition, so it should have something precise to say here. The thesis of this program is that it does: that there is one obstruction, living in

one cohomology group, that governs whether a directed system composes correctly, and that it specialises — with real content — to smart contracts, to agent orchestration, and to supply chains.

1.2 The gap

The pieces have existed separately. Directed containers are known to model stateful, navigable structures and to be equivalent to small categories (Ahman–Chapman–Uustalu; Shapiro–Spivak in the language of **Poly**). Zappa–Szép products are the classical algebraic device for gluing two structures that act on each other. Group cohomology supplies obstruction classes. What has been missing is the *fusion*: a single frame in which “these two directed systems compose” is literally a Zappa–Szép product in **DCont**, in which the obstruction to that product is a specific H^2 class, and in which the same class, read in an application, is the re-entrancy bug or the orchestration deadlock. That fusion, and its consequences, is what this program supplies.

1.3 The result, in one line

Throughout, “correctness” of a composition means: the composite directed container exists (its structure maps are well-defined and satisfy the directed-container laws) and computes what the two factors jointly specify. Our organising claim is:

A directed system composes correctly if and only if a lifting condition (L) holds and an obstruction class $[\omega] \in H^2$ vanishes.

The two halves are independent and have different characters. (L) is a first-order, degree-one “the quotient exists” condition; $[\omega] = 0$ is a degree-two “the pieces reassemble” condition. The program’s theory pillar (§2) makes both precise and proves the biconditional in the cases we have closed; the applications pillar (§3) exhibits $[\omega]$ as a concrete, sometimes machine-checked, invariant of real systems; the formalisation pillar (§4) verifies the arithmetic core of $[\omega]$ in Lean.

1.4 Method

The engine is the equivalence **DCont** $\simeq$ **Cat** (§2.1). It lets us move a question about composing directed systems into a question about welding two categories, where the classical apparatus — Zappa–Szép products, the Lyndon–Hochschild–Serre spectral sequence, group cohomology — is available and sharp. The obstruction $[\omega]$ is the class of a non-split extension; the lifting condition (L) is the existence of a suitable transversal. Applications are obtained by *specialising the base*: an agent orchestration, a contract’s storage, a supply graph each present as a particular directed container, and the general obstruction computes to the domain’s known failure mode.

1.5 Consequences and scope

The payoff is a design principle: to certify a composite directed system, check (L) and compute $[\omega]$. Where $[\omega]$ lands in a finite group — as it does for the re-entrancy model, where $[\omega] = \varepsilon \in \mathbb{F}_2$ — this is a decidable check, and we have machine-checked the arithmetic. This is what “compositional correctness has economic consequences” means concretely (§5).

Honest scope. This is a synthesis, not a new theorem. It introduces no new result; it maps existing ones. Three gaps are stated plainly rather than buried:

- (i) *Tax computation*, named in the grant as an application, has no artifact yet (§3).

- (ii) The representability wall $\mathcal{E} = 0$ (Conjecture 6.2 / Gap- S) is [open]; the September 2026 module reduction reframed it but did not close it (§2.7).
- (iii) Nothing is externally published; the strongest grade is Rick-refereed (§6).

1.6 Outline

§2 is the theory pillar: the equivalence chain, Zappa–Szép composition, the obstruction $[\omega]$, change of base, the full-faithfulness theorem, the H^2 -cluster, and the open wall. §3 is the applications pillar. §4 is the formalisation pillar. §5 states the impact. §6 is a consolidated status ledger, and §7 the roadmap.

2 Theory

2.1 The equivalence chain

Definition 2.1. A *container* is a pair (S, P) with S a set of *shapes* and $P: S \rightarrow \mathbf{Set}$ an S -indexed family of *positions*; it denotes the functor $\llbracket S, P \rrbracket X = \sum_{s \in S} X^{P(s)}$. A *directed container* adds to (S, P) operations making each fibre navigable — a “root” position, a way to move along a position, and a way to compose moves — subject to laws.

Theorem 2.2 (Equivalence chain). *There is a chain of equivalences*

$$\mathbf{DCont} \simeq \mathbf{Comon}(\mathbf{Poly}) \simeq \mathbf{Cat},$$

identifying directed containers with comonoids in $(\mathbf{Poly}, \triangleleft)$ and with small categories. Under it a directed container is a small category: shapes are objects, positions are morphisms out of an object, the root is the identity, and composition of positions is composition of morphisms. [external identification established; assembled chain in-progress]

The equivalence $\mathbf{Comon}(\mathbf{Poly}) \simeq \mathbf{Cat}$ is Shapiro–Spivak’s established $\mathbf{Cat}^\# = \mathbf{DCont}$ identification in $\mathbf{Poly}$; we take it as given and build on it. The chain is the program’s convergence hub: four independent lines of work (state liftings, holonomy, orchestration, obstruction theory) all land on the single statement $\mathbf{DCont} \simeq \mathbf{Cat}$. It is the reason a theorem proved about categories transports to a statement about directed systems, and conversely. The convergence fronts are individually [proved] (for instance, state liftings $\cong$ categorical completeness); the fully assembled and formalised chain as our own artifact is recorded [in-progress] in the registry, and is the natural target for the next formalisation push.

2.2 Zappa–Szép composition and the condition $(L) \wedge (G)$

Definition 2.3. Given two categories (equivalently, two directed containers) C and D that act on each other, their *Zappa–Szép product* $C \bowtie D$ is the category whose morphisms are pairs composed by a mutual “straightening” rule: D acts on C and C acts on D , and the two actions interleave. It generalises both the direct product (no interaction) and the semidirect product (one-sided action).

Theorem 2.4 (Pairwise Zappa–Szép criterion). *A category K decomposes as a Zappa–Szép product $K \cong C \bowtie D$ of two subcategories if and only if two conditions hold:*

(L) a lifting/factorisation condition, and (G) a global cocycle condition.

[proved]

This is the composition theorem in categorical form. Because $\mathbf{DCont} \simeq \mathbf{Cat}$, it is simultaneously a theorem about welding two directed systems. (L) says every morphism of K factors uniquely through the two factors (the “straightening” is well-defined); (G) says the resulting factorisation is globally coherent. The name of the game in the rest of the theory is: (L) is usually easy to arrange, and (G) is where the content lives — so we ask what (G) is.

2.3 The obstruction $[\omega] \in H^2$

Theorem 2.5 (The global condition is a cohomology class). *The global condition (G) is the vanishing of an obstruction class:*

$$(G) \iff [\omega] = 0 \in H^2,$$

where $[\omega]$ is the class of the extension welding the two factors. Holonomy composition realises the Zappa–Szép bridge, and trivial holonomy forces collapse (the State $\leftrightarrow$ Belief proposition). [peer-reviewed (holonomy–ZS bridge); proved (obstruction is H^2)]

This is the technical heart. Combining Theorems 2.4 and 2.5 yields the program’s slogan,

$$\text{correct composition} \iff (L) \wedge [\omega] = 0 \in H^2$$

The holonomy–Zappa–Szép bridge — that the composite’s holonomy is exactly the obstruction to a splitting — is [peer-reviewed]. Two further structural facts sharpen the picture: the emergent-holonomy count is a degree-zero invariant (H^0), with the higher Mackey/Shapiro data in H^n [proved]; and there are genuinely two distinct $[\omega]$ -sites that are *not* related by isotropy restriction [proved], so the obstruction is not an artifact of one presentation.

2.4 Change of base: when enrichment is free

Theorem 2.6 (Self-enriched change of base). *For a monad/comonad M used to change the base of the container construction, the induced functor is full and faithful if and only if $M = \text{Id}$. [proved] [Lean-verified]*

The lesson is a negative one worth stating: enriching the base — carrying extra effect structure M along the container construction — buys expressive power but, for $M \neq \text{Id}$, *costs* full-faithfulness; the extension is no longer a conservative embedding. This delimits how far the clean theory travels when one moves from **Set** to a richer base such as **Vec**, and it was independently machine-checked.

2.5 Full-faithfulness is unit-connectedness

Theorem 2.7 (Flagship, T1). *The container denotation functor $\llbracket - \rrbracket$ is full and faithful if and only if the monoidal unit is connected — not, as one might guess, if and only if the base is extensive. [proved]*

T1 is the program’s cleanest “the obvious guess is wrong” result. It was natural to expect that full-faithfulness of the container semantics tracked extensivity of the base; it does not. It tracks connectedness of the monoidal unit, a strictly different and sharper condition. This is the kind of result that a formalisation makes trustworthy and a synthesis exists to highlight.

2.6 The H^2 -cluster: holonomy and prunability

Theorem 2.8 (Two edges of one spectral sequence). *On $\mathbf{DCont} \simeq \mathbf{Cat}$ there are two a priori different degree-two obstructions — the Zappa–Szép holonomy $[\omega]$ and Ferri prunability — and they do not decompose as a direct sum. Instead they are the two edges of the Lyndon–Hochschild–Serre spectral sequence of the welding extension: $[\omega]$ is a base class and prunability the transgression d_2 , coupled by $d_2(\varphi) = \varphi_*[\omega]$. A clean splitting $H^2 = (\text{base}) \oplus (\text{fibre})$ holds if and only if $[\omega] = 0$. [proved; supporting fibre-edge kernel Lean-verified]*

An earlier conjecture that the cluster splits as a product was *refuted*; the correct statement is the spectral-sequence coupling above. We record this because the refutation is itself informative — it says $[\omega]$ and prunability are not independent knobs, they are linked by a differential, and the link degenerates exactly when the base obstruction vanishes. A third invariant, the direction-functor $H^2_{\mathbf{Cat}_B}$, was placed in the cluster: for a connected groupoid it reduces to $H^2(\pi; M)$ of the vertex group, and it is orthogonal to prunability [proved]. Prunability separates from $[\omega]$ in general (the full loop bundle always has $[\omega] = 0$) [proved].

2.7 The open wall: representability, Conjecture 6.2 / Gap-S

One question is genuinely open, and the program’s honesty depends on saying so clearly.

Conjecture 2.9 (6.2 / Gap-S). *For the $\mathbf{Vec}$ -admissibility question, the natural-transformation functor $F = \text{Hom}(E, \bigoplus_{\mathbb{N}} -)$ is projective in the relevant additive functor category (equivalently, a certain comparison map ι_* is injective). [open]*

The September 2026 module reduction is real progress that stops short of a proof. It transfers the question to modules over $S = \text{End}_k(E)$, computes the residual group $\text{Nat}(Q, Q)$ up to an Ext^1 wall, and — decisively — shows the *Baer–Specker obstruction is void*: the Specker identity is consistent with freeness, so no product-vs-sum argument can manufacture an exotic map. That explains why every attempted exotic construction failed and reframes the residual question ($\mathcal{E} = 0$) as one about a countable product of non-finitely-generated projectives over S . It is [proved] *as a reduction*; the wall itself, $\mathcal{E} = 0$, is [open]. We flag it here so no reader mistakes the reduction for a resolution.

3 Applications

Each application is obtained by specialising the base of the general theory. The pattern is always the same: present the domain object as a directed container, and the general obstruction $[\omega]$ computes to the domain’s known failure mode.

3.1 Agent orchestration is a Zappa–Szép product

Composing two agent orchestrations is literally the Zappa–Szép weld of §2.2: each agent is a directed container, and orchestrating them interleaves their control exactly as the mutual action of $C \bowtie D$. Correct orchestration is then $(L) \wedge [\omega] = 0$; a deadlock or a shared-state corruption is a non-vanishing $[\omega]$. [proved]

3.2 Re-entrancy is a failed distributive law, $[\omega] = \varepsilon \in \mathbb{F}_2$

A smart contract’s storage and its call structure are two directed systems that must compose via a distributive law. Re-entrancy is precisely the failure of that law, and the obstruction lands in a two-element group: $[\omega] = \varepsilon \in \mathbb{F}_2$, with $[\omega] = 0$ the safe contract and $[\omega] = 1$ the exploitable one. This is the sharpest specialisation in the program — a finite, decidable obstruction — and its arithmetic is [Lean-verified]. The standalone note packaging this result is [peer-reviewed] (Rick, 2026-09-03). Security audits are, in this language, the search for a failed distributive law.

3.3 Tool-calling protocols are directed containers

A tool-calling protocol (of the MCP kind used by agent harnesses) is a directed container: the protocol is the container, the harness is its comonad, the cofree construction is the execution trace, and multi-agent interleaving is the Zappa–Szép/distributive-law weld. Protocol correctness is again $(L) \wedge [\omega] = 0$, with $[\omega] = \varepsilon \in \mathbb{Z}/2$ (Lean). This is the most recent application and is written for an external audience (ACT/**Poly**); it imports Theorem 2.5 rather than reproving it. [proved] $[[\omega] = \varepsilon$ Lean-verified]

3.4 Supply chains

A supply graph with local reconciliation at each node is a directed container; global reconciliation is the vanishing of the composite obstruction. This is established on worked examples. [computed]

3.5 Genetic-algorithm and linear-attention composition

The container framework analyses compositional structure in learned systems: linear attention is a $\odot$ operation at fixed context, and the depth of a stack is not concentrated in the attention layer (the “stack = one matrix” guess is refuted). The **Vec**-based container theory (biproduct collapse, extensivity crux) is the setting. [proved] for the structural claims.

Gap. *Tax computation* is named in the grant as a target application. It has *no artifact* in the program to date. It is a genuine opportunity — the compositional structure of a tax code (rules composing, with global consistency conditions) fits the frame — but nothing has been built, and this document does not pretend otherwise.

4 Formalisation

The program’s Lean 4 development comprises on the order of two dozen registry nodes, mostly Mathlib-free core constructions, verifying the load-bearing arithmetic of the theory. The most important cluster is the H^2 engine trio, which mechanises the computation of the obstruction class:

- **cyclic-h2-gcd-engine** — the cyclic H^2 gcd computation. [Lean-verified]
- **cyclic-h2-norm-engine** — the cyclic norm/kernel computation. [Lean-verified]
- **fibre-h2-z2-engine** — the fibre-edge kernel, $H^2(\mathbb{Z}/2; \{0, 2\}) = \mathbb{Z}/2$, sorry-free. [Lean-verified]

Alongside these: the re-entrancy witness $[\omega] = \varepsilon$ is machine-checked (§3.2); change of base ($M = \text{Id}$, Theorem 2.6) is Lean-verified; the free-monad grafting/unit laws are verified sorry-free; and the first composition adjunction $F_q \dashv (-) \triangleleft q$ over **Vec** is verified with its naturality as an independent probe. These are the pieces of the theory whose correctness is not a matter of my say-so.

5 Impact

The unifying claim of the program is economically legible: *compositional correctness has economic consequences exactly where composition fails silently*, and the program pinpoints that failure as a single, often-computable obstruction.

- **Smart contracts.** The DAO-class re-entrancy bug is $[\omega] = 1 \in \mathbb{F}_2$. The obstruction is finite and decidable; auditing is checking $[\omega]$. (§3.2, peer-reviewed and Lean-verified.)
- **Agent orchestration.** Deadlock and shared-state corruption in a multi-agent system are non-vanishing $[\omega]$; correct orchestration is $(L) \wedge [\omega] = 0$. (§3.1, proved.)
- **Supply chains.** Global reconciliation failure despite local consistency is a non-vanishing composite obstruction. (§3.4, computed.)

The design principle for a practitioner is compact: **to certify a composite, check (L) and compute $[\omega]$** . Where $[\omega]$ lives in a finite group this is a decidable, and now machine-checked, procedure.

6 Status ledger

The following table is the honest, consolidated map. Grades are taken from the proof registry; where a topic spans several nodes the strongest relevant grade is shown, with weaker or open components noted.

Result	Grade	Artifact
<i>Theory</i>		
Equivalence chain DCont $\simeq$ Comon(Poly) $\simeq$ Cat	in-progress (fronts proved; ext. node given)	convergence-hub; reg. equivalence-chain
Pairwise ZS criterion $(L) \wedge (G)$	proved	reg. pairwise-zs
$(G) \iff [\omega] = 0 \in H^2$; holonomy-ZS bridge	peer-reviewed	reg. holonomy-composition-zs-bridge
Emergent holonomy $= H^0$; Mackey/Shapiro H^n	proved	reg. emergent-holonomy-is-ext-tower
Two $[\omega]$ -sites distinct (not isotropy)	proved	reg. two-omega-sites
Change of base f&f $\iff M = \text{Id}$	proved, Lean	change-of-base; reg. change-of-base
T1: f&f $\iff$ unit connected	proved	which-functors-are-containers; reg. fullness-unit-connectedness
H^2 -cluster = LHS two edges (product refuted)	proved (fibre edge Lean)	h2cluster-two-edges-spectral-sequ
Direction-functor $H^2_{\text{Cat}_B} = H^2(\pi; M)$	proved	reg. groupoid-zs-obstruction
Prunability $\perp [\omega]$ (separation)	proved	reg. quiver-skew-brace-zs
Representability wall $\mathcal{E} = 0$ (Conj. 6.2/Gap- S)	open (reduction proved)	mcontainers-codensity-polynomial proof 2026-09-20-module-reduction
<i>Applications</i>		

Result	Grade	Artifact
Orchestration = ZS product	proved	containers-for-orchestration
Re-entrancy = $[\omega] = \varepsilon \in \mathbb{F}_2$	peer-reviewed, Lean	reentrancy-machine-checked-obstru
MCP protocols = directed containers	proved ($[\omega]$ Lean)	tool-calling-protocols-directed-c
Supply chains	computed	harness-is-a-worker
GA / linear-attention composition	proved (structural)	reg. supply-chain-zs vec-attention-unification; workers-state-via-tensor
Tax computation	no artifact	—
<i>Formalisation</i>		
H^2 engine trio (gcd, norm, fibre- $\mathbb{Z}/2$)	Lean-verified	reg. cyclic-h2-gcd/norm, fibre-h2-z2
Re-entrancy witness $[\omega] = \varepsilon$	Lean-verified	reentrancy-machine-checked-obstru
Free-monad grafting / unit laws	Lean-verified	reg. free-monad-grafting
$F_q \dashv (-) \triangleleft q$ adjunction over Vec	Lean-verified	reg. left-adjoint-over-vec

The three gaps, restated. (i) Tax computation: no artifact. (ii) $\mathcal{E} = 0$ / Conjecture 6.2 / Gap- S : open. (iii) External publication: none; the strongest grade is Rick-refereed.

7 Conclusion and roadmap

The program has a spine and it holds: composition of directed systems is Zappa–Szép composition in $\mathbf{DCont} \simeq \mathbf{Cat}$, its correctness is $(L) \wedge [\omega] = 0 \in H^2$, the obstruction specialises to re-entrancy, orchestration, and supply reconciliation, and its arithmetic core is machine-checked. The slogan is not a simplification imposed after the fact; it is what the theorems say.

Three moves would most strengthen the grant narrative, in priority order:

1. **Close or cleanly frame the wall.** Either resolve $\mathcal{E} = 0$ (Conjecture 2.9) or write it up as a precise open problem about countable products of non-finitely-generated projectives over $S = \text{End}_k(E)$ — the reduction is already [proved] and publishable as such.
2. **Formalise the assembled equivalence chain.** Theorem 2.2 is the hub; upgrading it from [in-progress] to [Lean-verified] would anchor the whole theory pillar.
3. **Build the tax application, or drop it.** It is the one grant-named application with no artifact. The compositional structure of a rule system with global consistency conditions fits the frame; a worked example would close the gap, and if it does not fit, saying so is also progress.

Finally, the program has produced no externally published paper. The re-entrancy obstruction (finite, decidable, Lean-verified, economically vivid) and the tool-calling protocols note (written for ACT/**Poly**) are the two most venue-ready artifacts, and are the natural candidates for the first submission.

Sources

This synthesis cites internal artifacts by filename (papers in `projects/papers/`, proofs in `projects/proofs/`, registry nodes in `proofs/registry/`); the status ledger in §6 is the index. The external foundations the program builds on are: Ahman–Chapman–Uustalu on directed containers and Ahman–Uustalu on Zappa–Szép products from distributive laws; Shapiro–Spivak on **Poly** and the identification $\mathbf{Cat}^\# = \mathbf{DCont}$; and Niu–Spivak on Dirichlet convolution as Day convolution. Full bibliographic detail for these lives in the program’s source ledger (`memory/reading/sources.json`).